Lekha Patel

Contact Information

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Research interests

Domain-informed Statistical and Scientific Machine Learning, Statistical computing, Theory and applications of stochastic processes, Bayesian nonparametrics. Applications: Climate and Atmospheric Science, Cybersecurity & Bioinformatics.

Education

- **Ph.D. Mathematical Statistics (Elected Presidential Fellow and Dean's Awardee),** ***Imperial College London*** October 2015 — August 2019. Advisors: Dr. Edward Cohen and Prof. Niall Adams.
  - Thesis title: *Modeling and Inference of Spatio-Temporal Processes in Single Molecule Localization Microscopy.*
- **BS, MSci Mathematics,** ***Imperial College London*****, First Class Honors (GPA 4.0)** ***summa cum laude*** **(top 2% of cohort)** October 2011 — June 2015

Experience

- **R&D Computer Science Principal Computational Statistician,** ***Scientific Machine Learning Department, Sandia National Laboratories, Albuquerque NM*** December 2022 — Present
  - Research and development of advanced computational and statistical learning algorithms for climate science, cybersecurity, mechanical systems.
  - Principal Investigator (PI) of multiple Laboratory Directed Research & Development (LDRD) and customer-focused (Department of Energy) projects, leading large multi-disciplinary teams (including engineers, program managers and university faculty) for cross-cutting experimental and theoretical research. Current projects involve leading a team to fine-tune and develop a scalable multimodal Large Language Model (LLM) for rare event quantification, developing novel atmospheric stochastic frameworks for enhancing climate change research, constructing generalized statistical learning and computing tools for high-dimensional datasets including graphical time series characterization and streaming anomaly-detection.
  - Principal Investigator (PI) of a multi-lab $5M Research & Development project to build and analyze digital twins of cyber-physical systems as required to enhance grid security, and climate change anomalies to the grid.
  - Leading large team of scientists, engineers and university faculty to develop a multi-tier system for monitoring fugitive methane emissions in large spatial-temporal areas across the United States.
  - Development, research and direct application of advanced probabilistic spatio-temporal models for extreme/rare events with application to climate forecasting, handing missing and approximate data, to national security problems such as grid integration, cyber security and reliability.
  - Disseminating research through publications and conference presentations.
  - Referee for journals Bayesian Analysis, Bioinformatics, Annals of Applied Statistics, Environmetrics, Nature Climate Change and Nature Communications Earth & Environment.

  - Invited Program Committee member (PC) for the AAAIs Conference on Artificial Intelligence (AAAI), NeurIPS Climate Change ML, International Conference on Learning Representations (ICLR), AIStats.
- **Research Adjunct Assistant Professor, *Department of Computing, University of Utah*** June 2022 — Present
  - PhD student advising.
  - Fostering collaborations through joint DOE Office of Science and NSF proposals.
  - Research collaborations in foundational research of Scientific Machine Learning.
- **Chief data scientist, *XGenomes Corp, Boston MA*** January 2022 — June 2023
  - Primary statistician and data scientist that developed biologically-informed statistical machine learning models for fluorescent microscopy.
  - Collaborated closely with experimenters and ML software developers.
- **R&D Mathematics Statistician, *Statistical Sciences Department, Sandia National Laboratories, Albuquerque NM*** March 2021 — December 2022
  - Development of advanced computational and statistical learning algorithms to climate and cyber-security analysis.
  - Principal Investigator (PI) of two Laboratory Directed Research & Development (LDRD), leading large multi-disciplinary teams for cross-cutting experimental and theoretical research for atmospheric modeling and cyber security.
  - Disseminated research through publications and conference presentations.
  - Served as a referee for journals Bayesian Analysis, Bioinformatics and Annals of Applied Statistics.
  - Invited Program Committee member (PC) for the AAAI Conference on Artificial Intelligence (AAAI-21,22,23,24), and other top Machine Learning conferences.
- **Postdoctoral researcher, *Sandia National Laboratories, Albuquerque NM*** January 2020 — March 2021
  - Statistical and computational development of spatio-temporal point process models to problems in statistical signal processing, image analysis and climate informatics.
  - PI of Exploratory Express to conduct foundational research on biological imaging modeling.
  - Disseminating research through publications and conference presentations.
  - Served as a referee for journals Bayesian Analysis, Bioinformatics and Environmetrics.
- **Principal Data Scientist/Machine Learning engineer, *Filtered Technologies, London, UK*** January 2019 — January 2020
  - Research and development of deep and convolutional neural networks for the automatic labeling and (multi-class) classification of written documents used to train employees of numerous companies in different skills.
  - Techniques: word embeddings, dimensionality reduction, sample amplification through data re-sampling and synonym embedding replacement via nearest neighbors.
  - Computational development in Python of the machine learning architecture for

use on internal platforms and collaborations with other software engineers.

Research associate, ***Ward-Ober Laboratory, Texas A&M, USA*** October 2016 — February 2017

- Research of statistical time series models used to describe molecular fluorescence in super-resolution microscopy, in collaboration with experimenters.
- Development of a novel estimation procedure to infer fluorescing parameters and the derivation of a distribution describing molecular stoichiometry.
- Image analysis and a computational implementation of this method conducted in Java.

- **Graduate Instructor** ***Imperial College London, London, UK*** February 2015 — October 2019
  - Instructor for Mathematics undergraduate courses:
    Stochastic Simulation, Probability and Statistics (I & II) and Mathematics for Chemists I. Teaching assistant for the summer school in Machine Learning held by the Imperial College Business school.
  - Exam marker for the courses:
    Stochastic Simulation, Time Series, Probability and Statistics (I & II), Statistical Modeling, Survival Models, Medical Statistics.
  - Exam setter for the Probability and Statistics II January 2016 exam; exam invigilator.
- **Private Tutor,** ***Imperial College London, London, UK*** May 2012 — January 2019
  - Strong experience in mentoring and tutoring Imperial College Mathematics students with the courses:
    Applied Probability, Time Series, Machine Learning, Games and Decisions, Credit Scoring and Data Science.
  - Experience tutoring high-school students in advanced Mathematics for univeristy preparation.

Publications

- **Patel, L.** and F. Liang, "Bayesian variable selection for dependent data". *In preparation for Journal of the American Statistical Association (2026+).*
- Riemer, N., R. D. Davis, **L. Patel**, M. West, and A. Ault, "Aerosols in the era of Machine Learning". *In preparation for Bulletin of the American Meteorological Society (2026+).*
- Zenker, J., C.A. Pattyn, B. Bentz, and **L. Patel**, "Generating well-characterized cloud droplets in an agile tabletop chamber". *Under Revision at Atmospheric Measurement Techniques (2025+).*
- Schwertner-Watson, E. and **L. Patel**, "On cost-constrained feature selection". *In preparation for Statistical Science (2025+).*
- McGeehan, D., F. G. W. Christensen, and **L. Patel**, "A two-step approach to cyber intrusion detection under computational constraints". *In preparation for a computing conference (2026+).*
- Christensen, F. G. W., D. McGeehan, and **L. Patel**, "Preference elicitation for multiple-criteria decision analysis with an application in cybersecurity". *In preparation for The American Statistician (2025+).*

- Mokhtar, S., **L. Patel** and F. Christensen "Copula-Based Anomaly Detection for Dependent Multi-Sensor Systems". *In preparation for Technometrics (2026+).*
- Verzi, S., et al., "Measuring image quality after intelligent adaptive noise filtering". *In preparation for Entropy (2026+).*
- Krofcheck, D., C. Ulmer, **L. Patel**, A. Naugle, and J. Ray, "Vision-language models for blast damage assessment: A proof-of-concept probabilistic framework using street-level imagery". *In preparation for PLoS One (2026+).*
- **Patel, L.**, and L. Damiano, "Controller-Augmented Hidden Markov Models: A Computational Framework for Constrained Sequential Inference". *In preparation for IEEE Transacations on Pattern Analysis and Machine Intelligence.*
- **Patel, L.**, R. Marsh, S. Cox and E.A.K Cohen, "Dynamically varying cluster processes in Single Molecule Localization Microscopy (SMLM)". *In preparation for JASA Applications and Case Studies (2025+).*
- **Patel, L.** and K. Shuler, "Cost constrained feature selection with the elastic net". *In preparation for Transactions in Machine Learning Research (TMLR).*
- **Patel, L.**, et al., "Trust-Aware Multimodal Data Fusion for Yield Estimation: A case study of the 2020 Beirut Explosion ". *In preparation for PNAS Nexus (2026+).*
- **Patel, L.**, and S. Hossain-McKenzie, "Probabilistic Digital Twins for Cyber-Physical Systems: Heterogeneous State Estimation with Anomaly Detection and Uncertainty Quantification". *Submitted to Informs Journal on Data Science (2026+).*
- **Saleh, E.**, S. Ghaffari, J. H. Curtis, **L. Patel**, P. A. Bosler, N. Riemer, and M. West, "Reconstructing the Aerosol State from Partial Observations with Generative Modeling". *Under second revision at Artificial Intelligence for the Earth Systems (2026+).*
- **Saleh, E.**, S. Ghaffari, J. H. Curtis, **L. Patel**, P. A. Bosler, N. Riemer, and M. West, "Compact Aerosol Size-Composition Representations Preserving CCN, Optical, and Ice Nucleation Properties". *Under revision at Aerosol Science & Technology (AS&T) (2026+).*
- Rodriguez, J., S. Guha, **L. Patel** and K. Shuler, "Integrative Learning of Dynamically Evolving Multiplex Graphs and Nodal Attributes Using Neural Network Gaussian Processes with an Application to Dynamic Terrorism Graphs". *Under second revision at Journal of American Statistical Association (2026+).*
- Pattyn, C.A., J.P. Zenker, K. Mondragon, M. Tezak, K. Westlake, N. Jackson, **L. Patel**, and J.B. Wright, "Sub-grid scale mixing of injected particulate and a cloud-like aerosol in a laboratory environment". *Under revision at Quarterly Journal of the Royal Meterological Society (2026+).*
- Díaz-Ibarra, O. H., S. G. Frederick, J. H. Curtis, Z. D'Aquino, P. A. Bosler, **L. Patel**, C. Safta, M. West, and N. Riemer "TChem-atm (v2.0): A Scalable Performance-Portable Framework for Multiphase Atmospheric Chemistry". *Geoscientific Model Development (2026) 19(3), 1281–1299.*
- Corneck-Willcox, E.A.K. Cohen, J. S. Martin, **L. Patel**, F. Sanna-Passino, and K. Shuler, "Simultaneous global and local clustering in multiplex networks with

covariate information". *In press at Journal of Complex Networks (2026).*

- **Patel, L.**, E. Gonzalez, R. Kamm, and A. Hill, "Probabilistic Radioisotope Identification with List-Mode Gamma Ray Data". *Data Science in Science (2025).*

- Warburton, P., J.P. Zenker, K. Shuler, and **L. Patel**, "Image masks of global ship tracks for NASA MODIS data products". *Scientific Data (2025).*
- Jacobs, P.M, **L. Patel**, A. Bhattacharya, and D. Pati, "Minimax Posterior Contraction Rates for Unconstrained Distribution Estimation on $[0,1]^d$ Under Wasserstein Distance". *Transactions on Machine Learning Research (TMLR) (2024).*
- McMichael, L. A., M. Schmidt, R. Wood, P.N. Blossey and **L. Patel**, "Exploring ship track spreading rates with a physics-informed Langevin particle parameterization". *Geoscientific Model Development (2024).*
- **Patel, L.** and J.P. Zenker,"Assessing the Design of Integrated Methane Sensing Networks". *Environmental Research Letters (2024).*
- Pattyn, C. A., J. Zenker, B. J. Redman, J. D. van der Laan, A. L. Sanchez, K. Westlake, **L. Patel**, B. Bentz, J. Wright, "Influence of NaCl concentration on the optical scattering properties of water-based aerosols". *Applied Optics (2023).*
- Hopwood, M., **L. Patel** and T. Gunda, "Classification of photovoltaic failures with hidden Markov modeling, an unsupervised statistical approach." *Energies (2022).*

- **Patel, L.** and L. Shand, "Toward data assimilation of ship induced cloud-aerosol interactions." *Environmental Data Science (2022).*
- Shlomovich, L., E.A.K. Cohen, N. Adams and **L. Patel**, "Parameter Estimation of Binned Hawkes Processes". *Journal of Computational and Graphical statistics (2022).*
- **Patel, L.** *et al.*, "Marked spatio-temporal modeling of fatal events with an application to terror insurgencies." *(ArXiv preprint (2021).*
- **Patel, L.** *et al.*, "Blinking Statistics and Molecular Counting in direct Stochastic Reconstruction Microscopy (dSTORM)". *Bioinformatics (2021), ISSN:1367-4803* `[Code]`
- **Patel, L.** *et al.*, "A hidden Markov model approach to characterizing the photo-switching behavior of fluorophores". *The Annals of Applied Statistics (2019), 13(3), 1397-1429. ISSN: 1932-6157.* `[Code]`

Conference proceedings

- Beltre, A. M., D.C. Crowder, R. Kasamsetty, J. Murdock, **L. Patel**, S. G. Sanchez, C.M. Siefert, R. Roche, and N. Winovich "Performance of Vision-Language Models in Blast Severity Analysis". *Submitted to HPAI4S'26: The Second Workshop on HPC for AI Foundation Models & LLMs for Science.*
- Hossain-McKenzie, S.S., M. Sahakian, A. Hahn, J. Robinson, C. Abate, K. Metcham, T.Gray and **L. Patel**, "Multi-Physics Digital Twin Environment Design and Implementation for Integrated Nuclear Reactor and Electric Power Systems". *Under revision at IEEE International Conference on Communications Workshops (ICC Workshops): Cybersecurity of Digital Twins and Interconnected*

Cyber Physical Systems (SecCPS).

- **Patel, L.**, and S. Hossain-McKenzie "Heterogeneous Time Series Anomaly Detection in Cyber-Physical Systems with VAEM-LSTMs". *Accepted at International Conference on Machine Learning and Applications ICMLA 2025.*
- Jacobs, P.M, A. Bhattacharya, D. Pati, J.M. Phillips, and **L. Patel**, "Estimation of Large Zipfian Distributions with Sort and Snap". *Accepted to AIStats 2025.*
- Warburton, P., K. Shuler, and **L. Patel**, "Deep Learning Aerosol-Cloud Interactions From Satellite Imagery". *In Proceedings of the AAAI Symposium Series (2023).*
- Pattyn, C. A. ***et al.***, "Development and Characterization of a Tabletop Fog Chamber at Sandia National Laboratories" *Proc. SPIE (2022).*
- **Patel, L.** and L. Shand, "Simulating cloud-aerosol interactions made by ship emissions." *Proc. JSM (2020).*
- **Patel, L.** *et al.*, "Assessing Extreme Value Analysis to predict rare events from the Global Terrorism Database." *Proc. JSM (2020).*
- **Patel, L.** and E.A.K. Cohen, "Bayesian filtering for spatial estimation of photo-switching fluorophores imaged in Super-resolution fluorescence microscopy". *Proc. Asilomar Conference on Signals, Systems, and Computers (2018). ISSN: 2576-2303.*

Technical reports

- **Patel, L.**, et al., "Templates for Risk-informed Assurance with Curvature Embeddings (TRACE)". *Sandia Report, Sandia National Laboratories (2025).*
- **Patel, L.**, et al., "SEQUential ANomaly Analysis (SEQUANA)". *Sandia Report, Sandia National Laboratories (2024).*
- **Zenker, J. P. and Patel, L.** et al., "Methane Integrated Monitoring and Measurement System Design". *Sandia Report, Sandia National Laboratories (2024).*
- **Patel, L.**, et al., "Pluminate: Quantifying aerosol injection behavior from simulation, experimentation and observations." *Sandia Report, Sandia National Laboratories (2023).*
- **Patel, L.**, D. Maes and S.M. Anthony, "Molecular counting in super-resolution microscopy: A statistical approach". *Sandia Report, Sandia National Laboratories (2021).*
- L. Shand, A. Staid, K. M. Larson, **Patel, L.**, E. L. Roesler, D. Lyons, S. Gray, J. Hickey and K. M. Simonson, "Local limits of detection for anthropogenic aerosol-cloud interactions". *Sandia Report, Sandia National Laboratories (2021).*
- **Patel, L.**, L. Shand, J.D. Tucker and G. Huerta, "Spatio-temporal modeling of extreme global terrorism events". *Sandia Report, Sandia National Laboratories (2021).*
- **Patel, L.**, "Modeling and Inference of Spatio-Temporal Processes in Single Molecule Localization Microscopy". *PhD thesis, Imperial College London (2019).*
- **Patel, L.**, "Bayesian Variable Selection for Dependent Data". *Research project report, Imperial College London (2015).*

- **Patel, L.** et al., "Extreme value analysis of financial time series". *Research project report, Imperial College London (2013).*

POSTDOCTORAL ADVISING

- Georgia Smits (Dec 2025 -) on *Streaming rashamon sets for online change-point detection* . Department of Statistics, University of Washington.
- Jorge Sebastian Salinas (Jan 2025 -) on *Stochastic differential surrogate modeling of atmospheric processes.* Combustion Research Facility, SNL.
- Efrain Humberto-Gonzalez (Nov 2024 - Sept 2025) on *Statistically learning radioisotope decay.* Cognitive & Emerging Computing, SNL.
- Michael Schmidt (October 2021 - October 2022) on *Particle methods for aerosol injection simulation.* Computational Science, SNL.

DOCTORAL THESIS ADVISING

- Garrett Allen (2027, expected) on *Streaming Intrusion Detection on Cyber Networks.* Department of Statistics, University of Washington (UW). Primary advisor with Tyler McCormick (UW).
- Erin Schwertner-Watson (2027, expected) on *Feature engineering in optimal experiemental design.* Scientific Machine Learning, SNL and Department of Mathematics and Statistics, University of New Mexico. Primary advisor with Fletcher Christensen (UNM).
- Peter Jacobs (graduated in 2025) on *Bayesian nonparametrics and applications.* Statistical Sciences, SNL and Department of Computing, University of Utah. Primary advisor with Jeff Phillips (U Utah), Lajos Horvath (U Utah), Anirban Bhattacharya (TAMU) and Debdeep Pati (TAMU).
- Jason Torchinsky (graduated in 2023) on *Kalman filtering for solving supersaturation of plume injections.* Computational Science, SNL and Department of Mathematics, University of Wisconsin-Madison. Primary advisor: Sam Stechmann.
- Michael Hopwood (graduated in 2022) on *Statistical analysis of PV system failures.* Photovoltaics & Materials Technology, SNL and Department of Statistics and Data Science, University of Central Florida.
- Leigh Shlomovich (graduated in 2022) on *Parameter Estimation of Aggregated Hawkes Processes* (2019). Department of Mathematics, Imperial College London. Primary advisors: Ed Cohen and Niall Adams.

MASTER'S THESIS ADVISING

- Kirara Kura (March 2024 -) on *Streaming Intrusion Detection on Cyber Networks.* Department of Statistics, University of Washington. Primary advisor with Tyler McCormick (UW).
- Pierce Warburton (June 2022 -) on *Physics-informed Machine Learning of eddy diffusivity in atmospheric processes.* Scientific Machine Learning, SNL. Principal mentor.
- Petros Peppas (2019) on *Deep learning for Super-resolution microscopy.* Department of Mathematics, Imperial College London. Primary advisors: Ed Cohen and Seth Flaxmann.
- Elad Mazor (2018) on *Machine learning for Super-resolution microscopy.*

Department of Mathematics, Imperial College London. Primary advisor with Ed Cohen and Seth Flaxmann.

Other mentoring

- Women in Computing and Informational Sciences (W-CIS) (January 2023 - Present) *Specific SNL mentorship for early career women at Sandia.*
- Emily Kemp (March 2021 - Sept 2023) *general SNL mentorship.* Statistical Sciences, SNL.
- James Hickey (February 2021 - September 2021) on *Ship track identification from satellite imagery.* Statistical Sciences, SNL.
- Leigh Shlomovich (2016) on *Bayesian cluster analysis.* Department of Mathematics, Imperial College London.

Honors and Awards

- Up and Coming Sandia National Laboratories Innovator award (1400 director chosen) April 2024
- SNL spot award for providing leadership feedback April 2024
- SNL spot award for creating Women in computing and informational sciences group October 2023
- SNL solo Employee Recognition Award for diversity and inclusion March 2022
- SNL spot award for team building November 2021
- SNL spot award for 2021 SNL recruiting efforts November 2021
- SNL 2020 Postdoctoral showcase participation spot award June 2021
- SNL 2020 Postdoctoral showcase audience's choice spot award January 2021
- SNL spot award for successful LDRD proposal October 2020
- Imperial College PhD President's Scholarship October 2015 — August 2019
- Imperial College Dean's award May 2018
- Imperial College Sparx Prize May 2014

Grants

- Defense Nuclear Nonproliferation Research & Development (NNSA), *TANGO: Twinned Analysis of Nuclear reactor & Grid Operations*, Principal Investigator (PI) & Technical Lead, for $6,300,000, with Idaho National Laboratory and Oak Ridge National Laboratory November 2024 — October 2027
- Laboratory Directed Research & Development, *PARADIGM: Principled Approaches to Research and Development of Innovative Generative Models Team Member*, Methods Thrust & Technical Lead, for $12,000,000, October 2024 — September 2027
- Laboratory Directed Research & Development, *Bridging aerosol representations across scales with physics-informed statistical learning*, Principal Investigator with Pete Bosler (co-PI) & Technical Lead, for $2,300,000, with University of Illinois Urbana Champaign and University of Michigan. October 2023 — September 2026
- Laboratory Directed Research & Development, *SEQUANA: Sequential Anomaly Analysis*, Principal Investigator & Statistical Learning Technical Lead, for

$1,450,000, with University of New Mexico, Texas A&M University and University of Washington. October 2021 — September 2024

- Laboratory Directed Research & Development, *Pluminate: Quantifying aerosol injection behavior from simulation, experimentation and observations*, Principal Investigator & Statistics Technical Lead, for $1,070,000, with University of Washington. October 2021 — September 2023
- Laboratory Directed Research & Development, Principal Investigator & Technical Lead, for $89,000. October 2020 — September 2021

PROPOSALS

- DOE Advanced Scientific Computing Research, Office of Science, Statistical Machine Learning Technical Lead, for $6,000,000. Full proposal submitted, May 2024.
- Laboratory Directed Research & Development, Principal Investigator, *Domain-informed semi-parametric statistical learning under model misspecification* for $1,115,000. Full proposal submitted, May 2024.
- Defense Nuclear Nonproliferation (DNN) R&D (NA-22), Principal Investigator, *TANGO: Twinned Analysis of Nuclear reactor and Grid Operations* for $4,800,000 with Idaho National Laboratory and Oak Ridge National Laboratory Full proposal submitted after down-selection, April 2024.
- Laboratory Directed Research & Development, co Principal Investigator, *Bridging aerosol representations across scales with physics-informed statistical learning* for $1,944,000. Awarded, October 2023.
- DOE Biological and Environmental Research, Office of Science, Statistical Machine Learning Technical Lead, for $7,000,000. Full proposal submitted, not awarded, September 2023.
- DOE Advanced Scientific Computing Research, Office of Science, Statistical Learning Technical Lead, for $5,000,000. Full proposal submitted, not awarded, September 2023.
- Laboratory Directed Research & Development, Principal Investigator, *Unsupervised anomaly detection of extreme events with nonparametric time-varying quantile discriminators* for $1,450,000. Awarded, October 2022.
- Laboratory Directed Research & Development, Principal Investigator, *Quantifying aerosol injection behavior from large-scale satellite imagery using statistical modeling and machine learning* for $1,070,000. Awarded, October 2022.
- DOE Advanced Scientific Computing Research, Office of Science, Principal Investigator. Sandia down-selected, pre-proposal submitted, full proposal not requested, April 2021.
- Laboratory Directed Research & Development, Principal Investigator, *Molecular counting in super-resolution microscopy: A statistical approach* for $89,000. Awarded, Oct 2020.

RECENT INVITED TALKS

- Invited talk at ESCO 2024 - 9th European Seminar on Computing on *Physics-Informed Machine Learning of Aerosol-Cloud Interactions for Climate*

Geoengineering
Pilsen, Czech Republic June 2024

- Invited talk and minisymposium organizer at SIAM Uncertainty Quantification conference on *Physics-Informed Machine Learning of Aerosol-Cloud Interactions*
Trieste, Italy February 2024
- Invited talk at John's Hopkins Applied Physics Laboratory (APL) on *Learning Marine Cloud Brightening Dispersion for Climate Geoengineering*
Online February 2024
- Invited keynote talk at Women In Cybersecurity webinar on *Intrusion Detection in Cyber security*
Online March 2023
- Invited talk at Department of Statistics, University of Washington on *Flexible Hawkes Processes*
Seattle February 2023
- Invited talk at Department of Mathematics, University of Utah on *Flexible Hawkes Processes*
Salt Lake City October 2022

Service

- Created and lead Women in CIS peer-mentoring group (December 2023 - Present) for women researchers at the laboratory.
- Referee for Bayesian Analysis, Environmetrics, Environmental Data Science, Bioinformatics, Bioinformatics Advances, AAAI, NeurIPS, ICML.
- ASA New Mexico Chapter, Imaging and Nonparametrics Section.
- SIAM Activity Group on Data Science.
- SIAM group on Uncertainty Quantification.

Computing programs

- Proficient: Python, Julia, Matlab, R, C, C++, Latex
- Working knowledge: Java, Shell scripting

Operating systems

Linux, Mac OS, Windows.

Languages

English, Spanish.

References

Available upon request.